Maurice D. Hanisch

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Education

California Institute of Technology — *Pasadena, USA*
Ph.D. in Computing and Mathematical Sciences, GPA: 3.94 — *09/2025 – 06/2029*
- Research in **AI for quantum chemistry and materials** (neural operators, equivariant GNNs); advised by Prof. **Anima Anandkumar**.

ETH Zurich — *Zurich, Switzerland*
M.Sc. in Physics, GPA: 5.77/6.00 (Top 12%) — *09/2022 – 05/2024*
- Specialized in **quantum information theory**; earned 120+ credits (90 required), adding math and ML courses.

Ludwig Maximilian University of Munich — *Munich, Germany*
B.Sc. in Physics, GPA: 1.28/1.00 (Top 10%) — *10/2018 – 09/2022*
- Emphasis on **theoretical physics** and **mathematical methods** for quantum systems.

Experience

California Institute of Technology — *Pasadena, USA*
AI for Science Graduate Researcher, A. Anandkumar's Group — *09/2025 – Present*
- Designed and developed (co-first author) an SE(3)-equivariant **Fourier neural operator** that learns the Kohn-Sham map, replacing diagonalizations in **density functional theory (DFT)** for quasi-linear scaling (**runtime exponent 1.03 vs. 3.37** for Quantum ESPRESSO).
- Built the training stack in **PyTorch Lightning** (**multi-GPU DDP**, Slurm, L40S/H200/B300) and scaled to **8,250 atoms on one B300 GPU**.
- Modified **Quantum ESPRESSO** (Fortran/MPI) and built the full data pipeline, generating **59,133 potential-density pairs** (8,504 structures).

Bloomberg CTO Office — *New York, USA*
Security Research Intern, J. Vanegue's Team — *06/2025 – 09/2025*
- Found attack vectors against the **CHERI** secure CPU architecture; co-author of an **ACM CCS 2026** paper, a tier-1 security venue.

California Institute of Technology — *Pasadena, USA*
Machine Learning Research Intern, A. Anandkumar's Group — *09/2024 – 05/2025*
- Led a benchmark **retraining 7 state-of-the-art molecular models** (e.g., EquiformerV2) on **charged and open-shell** molecules.
- Built a differentiable **xTB** (tblite) pipeline feeding **orbital-based quantum features** to **OrbitAll**'s SE(3)-equivariant GNNs.

IBM Research Europe — *Zurich, Switzerland*
Quantum Computing Research Intern, S. Woerner's Group — *09/2023 – 05/2024*
- Built a Python/C++ soft-decoding pipeline processing **billions of measurement records** for one of IBM's then-largest QEC experiments.
- Ran the full experimental pipeline in **Qiskit** on a 127-qubit IBM Quantum device (error-correcting repetition codes up to distance 51), showing that soft (analog) readout decoding raises the **repetition-code threshold by 24%** and cuts logical error rates by up to **30**×.

ETH Zurich — *Zurich, Switzerland*
Quantum Hardware Research Assistant, J. Home's Group — *06/2023 – 08/2023*
- Engineered and **experimentally validated** ion-trap potentials for a tunable interaction between **GKP qubits** on trapped-ion hardware.

Max Planck Institute of Quantum Optics — *Garching, Germany*
Quantum Algorithms Undergraduate Researcher, I. Cirac's Group — *04/2022 – 09/2022*
- Proved that **bosonic Gaussian quantum circuits** can be implemented with **exponentially fewer qubits** than a direct encoding.

Selected Publications

- Learning the Kohn-Sham map with neural operators for quasi-linear scaling density functional theory
  D. Khan, **M.D. Hanisch***, N. Argatoff, E. Xie, S. Sharma, A. Anandkumar. Submitted to **Science***
- OrbitAll: A Unified Quantum Mechanical Framework for Deep Learning Across Diverse Molecular Regimes
  B.S. Kang, V.C. Bhethanabotla*, A. Tavakoli, **M.D. Hanisch**, . . . , A. Anandkumar. Under review at **Nature Computational Science***
- Soft information decoding with superconducting qubits
  ***M.D. Hanisch**, B. Hetényi, J.R. Wootton. **APS Open Sci.** 1, 000019 (2026)*

Awards and Competitions

- **Kortschak Fellowship**, California Institute of Technology (2025): two years of full Ph.D. funding, awarded to 5 incoming CMS students.
- **Best Poster Award** (5 of 60+ posters), Gordon Research Conference on Computational Materials Science and Engineering (2026).
- **Hackathons** ($3,500+ in prizes): **1st/30** ETH QHack 2024 (*NVIDIA QAOA*) | **1st/12** LOQC 2023 (*EDF*) | **2nd/6** ETH QHack 2023 (*IQM*)

Skills

ML/deep learning:	PyTorch (Lightning), neural operators (FNO), equivariant GNNs (e3nn), transformers, LLMs, multi-GPU DDP
Scientific computing:	DFT/electronic structure (Quantum ESPRESSO, PySCF), xTB/tblite, ASE, RDKit, numerical linear algebra
Quantum computing:	Qiskit, quantum error correction, Hamiltonian simulation
Programming:	Python, C++, C, Fortran, Linux, Slurm/HPC, Git, Docker
Open source:	Merged PRs to neuraloperator (FNO), tblite (xTB), HyperQueue (HPC), cheribsd
Languages:	German (native), French (native), English (C1: IELTS 8/9), Italian (A1)
Interests:	Weightlifting, volleyball, calisthenics, surfing, chess